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XRA-GS: X-ray Attenuation-Aligned Gaussian Splatting for Sparse Tomographic View Synthesis

Abstract

*3D Gaussian Splatting has been extended to X-ray novel view synthesis, yet existing methods retain the gradient-driven densification rules from natural-image 3DGS without modification. Under sparse CT acquisition, these rules cause Gaussians to accumulate near high-contrast anatomical boundaries while interior tissues that determine X-ray attenuation remain under-represented. We present **XRA-GS**, an X-ray Attenuation-Aligned Gaussian Splatting framework that corrects this structural imbalance by redesigning the Gaussian evolution pipeline. XRA-GS introduces three components: a support-profile seeding strategy that initializes Gaussians from the coarse attenuation support estimated by FDK reconstruction; a geometry-aware pruning step that removes Gaussians concentrated near anatomical boundaries; and an adaptive density modulation step that applies position-dependent density refinement throughout training. Experiments on five organs under a sparse-view CT protocol demonstrate that XRA-GS achieves state-of-the-art novel view synthesis quality against six representative baselines from both visible-light sparse-view synthesis and X-ray CT methods.*

1. Introduction

Computed tomography (CT) is an essential imaging technique for noninvasively examining the internal structure of an object, and is widely used in clinical diagnosis, image-guided intervention, and industrial inspection [**feldkamp1984practical**; **andersen1984simultaneous**; **wang2008image**]. During a CT acquisition, an X-ray source rotates around the object while a detector records multi-angle attenuation measurements, from which cross-sectional images or projections at unseen angles can be recovered through tomographic reconstruction. Reducing the number of acquired views is highly desirable, since it directly lowers radiation dose, shortens scan time, and broadens applicability to patients who cannot sustain long, stable acquisitions [**li2023gloredi**; **wang2024reducing**]. In this work, we study *Sparse Tomographic View Synthesis*: from only a handful of X-ray projections, the model must predict the line-integrated attenuation observed at unseen acquisition angles, so that each synthesized projection remains consistent with the Beer–Lambert imaging geometry rather than merely visually similar to the inputs. A reliable solution supplies auxiliary views for anatomical localization, lesion-range inspection, and intraoperative planning under severely limited sampling, where conventional analytical reconstruction breaks down [**willemink2019evolution**].

Visible-light and X-ray imaging are governed by fundamentally different rendering equations, and this difference dictates where representation capacity should reside. In visible-light 3DGS, a pixel is formed by front-to-back α-compositing of an emission–absorption volume rendering integral [**kerbl20233d**; **max1995optical**]: the accumulated transmittance decays multiplicatively along the ray, so for opaque scenes the integrand is dominated by Gaussians near the first-hit surface while contributions from deeper Gaussians are progressively masked. The surface-clustered Gaussian layout that emerges under α-compositing is therefore a direct consequence of this transmittance-induced concentration, and is consistent with the surface-aligned geometry that surface-oriented 3DGS variants exploit explicitly [**guedon2024sugar**; **huang20242dgs**]. In X-ray imaging, however, the detector reading follows the Beer–Lambert law [**kak2001principles**]: after taking the logarithm, the recorded quantity becomes a linear, order-independent line integral of the attenuation coefficient along the ray, with no transmittance decay and no occlusion. Consequently, every Gaussian on the path contributes on equal footing, and capacity must be distributed *along the entire ray path* rather than collapsed onto any single layer. Existing X-ray Gaussian methods such as X-Gaussian, R2-Gaussian, X-Field, and DGR have correctly replaced α-compositing with a radiative line-integral renderer [**cai2024xgaussian**; **hu2024r2**; **xfield2025**; **dgr2025**], and sparse-view 3DGS pipelines such as FSGS, CoR-GS, and DNGaussian further compensate for missing observations through depth or co-regularization priors [**zhu2024fsgs**; **zhang2024cor**; **li2024dn**]. Yet all of them inherit a Gaussian evolution rule that was originally designed for surface-centric α-compositing [**kerbl20233d**; **roessle2024densification**].

Crucially, the issue is not gradient-driven densification itself. Splitting or cloning Gaussians where the projection-error gradient is large is a sound idea, and our method retains exactly this mechanism. The issue is that under a line-integral renderer, the residual gradient is still concentrated at high-attenuation-contrast interfaces, such as bone–soft-tissue boundaries, which almost coincide with the high-color-contrast interfaces that drive visible-light densification. A naïve port therefore keeps spawning Gaussians at anatomical boundaries even though the rendering equation now asks for capacity *along* the path. Under sparse views, this bias surfaces not as insufficient coverage but as a structural *capacity misallocation*: a substantial fraction of Gaussians is spent on boundary-driven redundancy, while interior path segments that determine the attenuation integral remain under-represented. The top row of Fig. 1 shows the natural fit between α-compositing and surface-clustered Gaussians; the bottom row shows that when the same evolution rule is reused under line-integral rendering, capacity still collapses to boundaries and leaves the deep path empty. Fixing this requires *augmenting*, not discarding, gradient-driven densification with attenuation-aligned decisions about where to seed, what to prune, and how to refine.

Motivated by this analysis, we propose **XRA-GS**, an X-ray Attenuation-Aligned Gaussian Splatting framework for Sparse Tomographic View Synthesis. Rather than appending auxiliary branches to a conventional densification backbone, **XRA-GS** keeps the gradient-driven backbone of standard 3DGS and adds three attenuation-aligned stages that act on *where* Gaussians are seeded, retained, and refined, as illustrated in Fig. 1. Support-Profile Seeding (SPS) anchors initialization to a coarse FDK attenuation support [**feldkamp1984practical**], so that Gaussians populate physically meaningful X-ray paths instead of starting from a contrast-biased prior; Geometry-aware Pruning (GAP) recycles Gaussians that accumulate near high-contrast anatomical interfaces during training, draining boundary-driven redundancy at its source; and Adaptive Density Modulation (ADM) applies position-dependent online refinement so that retained Gaussians carry interior attenuation rather than uniformly inflated density. Across diverse organs and view budgets, this attenuation-aligned evolution reallocates representation capacity from boundary redundancy to attenuation-relevant interior structures.

Our main contributions are summarized as follows:

- We identify *capacity misallocation* as the failure mode when sparse-view X-ray Gaussian methods inherit a visible-light evolution rule under a line-integral renderer: gradient-driven densification keeps clustering Gaussians at attenuation boundaries while interior path segments that determine the line integral remain starved of capacity.
- We propose **XRA-GS**, an X-ray Attenuation-Aligned Gaussian Splatting framework that augments standard gradient-driven densification with three coordinated attenuation-aligned stages—Support-Profile Seeding (SPS) for path-anchored initialization from coarse FDK support, Geometry-aware Pruning (GAP) for boundary-redundancy recovery, and Adaptive Density Modulation (ADM) for position-dependent online refinement—without modifying the X-ray renderer.
- Extensive experiments on multi-organ sparse-view CT show that **XRA-GS** delivers consistent gains in projection fidelity over state-of-the-art X-ray Gaussian and sparse-view 3DGS baselines, while reducing the number of Gaussians and training cost; ablations confirm that SPS, GAP, and ADM each contribute complementary capacity reallocation.

2. Related Work

2.1. Gaussian Splatting

3D Gaussian Splatting (3DGS) represents a scene with a set of anisotropic Gaussians and renders novel views in real time through differentiable tile-based rasterization [**kerbl20233d**]. Building on this explicit representation, subsequent work has expanded along several directions: surface-aligned variants that recover geometrically accurate meshes from Gaussians [**huang20242dgs**; **guedon2024sugar**]; dynamic and deformable extensions for time-varying scenes [**fourdgs2024**; **wu2024deformable**]; dense visual SLAM systems built on top of the Gaussian representation [**yan2024gsslam**]; generative pipelines that lift 3DGS into 3D content creation [**tang2024dreamgaussian**]; and compactness/efficiency redesigns that reduce the Gaussian count or stabilize rendering across scales [**lu2024compact**; **scaffoldgs2024**; **mipsplatting2024**]. Across all these directions, the underlying evolution rule remains the gradient-driven clone-and-split densification together with spherical-harmonics view-dependent color, both of which are tailored to dense-view, surface-reflectance scenes. When either assumption is violated—inputs become sparse, or the imaging model switches from surface radiance to line-integral attenuation—this rule starts to misallocate Gaussian capacity, and the next two subsections examine these two regimes in turn.

2.2. Sparse View Synthesis

Neural Radiance Fields (NeRF) recover a continuous volumetric field through coordinate MLPs and volume rendering [**mildenhall2021nerf**], and degrade noticeably when only a handful of input views are available. NeRF-side sparse-view methods approach this limitation from complementary angles, each adding a different external prior that has been empirically validated: depth supervision from sparse structure-from-motion points [**deng2022dsnerf**; **sparsenerf2023**], semantic consistency from pretrained vision encoders [**jain2021dietnerf**], and geometric/appearance regularization with frequency-domain annealing [**niemeyer2022regnerf**; **freenerf2023**]. Sparse-view 3DGS work explores a similarly diverse set of perspectives on the explicit Gaussian set: FSGS attributes degradation to insufficient coverage and augments densification with proximity-guided unpooling and depth priors [**zhu2024fsgs**]; CoR-GS treats it as unstable optimization and adds dual-field co-regularization with co-pruning [**zhang2024cor**]; DNGaussian injects geometry through depth normalization [**li2024dn**]; and DropGaussian counters overfitting via stochastic Gaussian dropout [**park2025dropgaussian**]. A complementary feed-forward line further moves geometry completion into large-scale pretraining [**charatan2024pixelsplat**; **chen2024mvsplat**; **depthsplat2025**; **liu2025monosplat**]. These different angles all yield empirical gains, which suggests that sparse-view degradation admits multiple compatible explanations rather than a single root cause. The next subsection turns to how that assumption itself is challenged under X-ray line integration.

2.3. Tomographic Novel View Synthesis

Sparse-view tomographic novel view synthesis aims to recover unseen X-ray projections, and the underlying density volume,

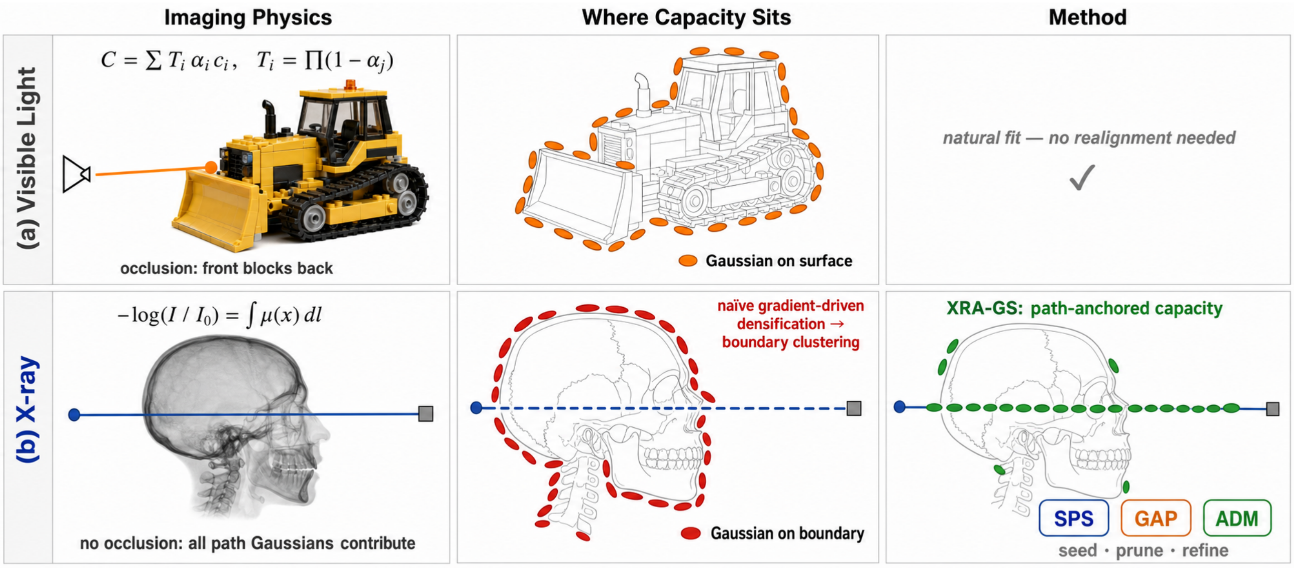

Figure 1: ***Visible-light vs. X-ray Gaussian rendering and the capacity gap they impose.*** *(a) Visible-light α-compositing concentrates capacity on visible surfaces, where front Gaussians occlude back ones. (b) X-ray line integrals weight every Gaussian along the ray equally, so naïve gradient-driven densification still clusters at attenuation boundaries;* ***XRA-GS*** *realigns the evolution rule through SPS, GAP, and ADM to anchor capacity along the path.*

from a small number of input projections [**wang2008image**; **willemink2019evolution**], in contrast to analytical (FDK) and iterative (SART) reconstructions that either suffer severe streak artifacts under sparse views or require many iterations [**feldkamp1984practical**; **andersen1984simultaneous**]. Learning-based methods fall into two families with complementary entry points. Implicit methods adapt neural attenuation fields to X-ray imaging—through self-supervised sinogram synthesis, hash-encoded acceleration, transformer-based structural modeling, or population-level encoder–decoder inference [**zang2021intratomo**; **guan2024naf**; **cai2024saxnerf**; **liu2025geometryaware**]. Explicit methods redesign Gaussians around X-ray attenuation, covering radiative kernels with X-ray rasterization, integration-bias correction with differentiable voxelization, and material-aware ellipsoids with explicit segment-length physics, alongside further extensions toward discretized voxels, artifact-suppression graphs, 4D dynamics, and slice-wise reconstruction [**cai2024xgaussian**; **hu2024r2**; **xfield2025**; **dgr2025**; **grgaussian2025**; **x2gaussian2025**; **layerbased2025**] (the last four target imaging regimes outside our sparse-view novel-view protocol and are omitted from the main comparison). Across both families, improvements have concentrated on the imaging equation or the kernel itself rather than on the evolution rule that decides where Gaussians accumulate, leaving sparse-view CT still bottlenecked by capacity misallocation inherited from visible-light densification. **XRA-GS** addresses this evolution rule directly, with SPS, GAP, and ADM aligning initialization, pruning, and refinement to the X-ray attenuation path.

3. Method

3.1. Preliminary

3.1.0.1. X-ray attenuation imaging. An X-ray detector pixel records the residual energy of a single ray after traversing the object, governed by the Beer–Lambert law. For a ray $\mathbf{r}$ with source intensity I_0 and attenuation field $\mu(\mathbf{x})$, the standard practice is to work in the logarithmic domain [**kak2001principles**], so that each pixel becomes a line integral of attenuation along the ray:

$$I(\mathbf{r}) = -\log \frac{I'(\mathbf{r})}{I_0} = \int_{\mathbf{r}} \mu(\mathbf{x})\,\mathrm{d}l. \tag{1}$$

Unlike α-compositing in visible-light rendering, this integral is linear and order-independent: every point along $\mathbf{r}$ contributes additively, with no occlusion between front and back materials.

3.1.0.2. Radiative Gaussian backbone. We represent the scanned object with a set of radiative Gaussians $\mathcal{G} = \{G_i\}_{i=1}^{N}$, following the recent line of explicit Gaussian-based tomographic representations [**cai2024xgaussian**; **hu2024r2**]. Each Gaussian G_i is parameterized by a center $\mathbf{p}_i$, a covariance Σ_i, and a non-negative density ρ_i, defining a local density field

$$G_i(\mathbf{x}) = \rho_i \exp\Big(-\tfrac{1}{2}(\mathbf{x}-\mathbf{p}_i)^{\top}\Sigma_i^{-1}(\mathbf{x}-\mathbf{p}_i)\Big), \tag{2}$$

and the global attenuation field is the sum $\mu(\mathbf{x}) = \sum_{i=1}^{N} G_i(\mathbf{x})$. Substituting this representation into Eq. (1) admits a differentiable X-ray rasterizer $\mathcal{R}$, so that for any acquisition geometry π_v a projection can be rendered as

$$\hat{I}_v = \mathcal{R}(\mathcal{G};\pi_v). \tag{3}$$

3.1.0.3. Problem statement. Given a sparse-view observation set $\mathcal{P} = \{(I_v, \pi_v)\}_{v=1}^{V}$ and a coarse FDK reconstruction V_{FDK}, our goal is to learn $\mathcal{G}$ in a per-case optimization setting so that target projections at unseen angles can be synthesized from only a few observed projections. The radiative backbone has already aligned the renderer with X-ray physics; what remains misaligned in this regime is the evolution rule that shapes $\mathcal{G}$ during training, where gradient-driven densification continues to concentrate Gaussians at high-gradient anatomical boundaries. **XRA-GS** realigns this evolution rule through three modules introduced below, while leaving the rasterizer and the differentiable voxelizer of R2-Gaussian [**hu2024r2**] unchanged.

3.2. Overall Framework

As illustrated in Figure 2, **XRA-GS** instantiates the attenuation-aligned evolution pipeline as three modules acting at three distinct stages of optimization: SPS replaces standard initialization by seeding Gaussians along the FDK attenuation support; GAP acts inside the densification loop and reclaims locally redundant Gaussians produced by gradient-driven growth; ADM acts on retained Gaussians and refines their density values using position-dependent spatial context. At inference, the rasterizer remains identical to the radiative backbone, so the added cost is concentrated in training.

3.3. Support-Profile Seeding (SPS)

Initialization strongly conditions the optimization trajectory of radiative Gaussians, yet standard 3DGS initializers transfer poorly to X-ray tomography: uniform sampling ignores the strong contrast between air, soft tissue, and bone, while SfM point-cloud initialization is unstable under gray-value overlap and weak local texture in X-ray projection [**cai2024xgaussian**]. The sparse-view CT literature has consistently treated the coarse FDK volume V_{FDK} as carrying reliable foreground support and low-frequency attenuation layout, even when its fine detail is corrupted by streak artifacts [**wang2022rfinet**; **li2023gloredi**; **li2025threedgrct**; **hu2024r2**]. We exploit this signal as a sampling prior rather than as a target volume. Concretely, over the voxel grid Ω underlying V_{FDK} we sample seed positions from a density-weighted mixture distribution

$$q(\mathbf{x}) = \alpha \frac{1}{|\Omega|} + (1-\alpha)\frac{V_{\text{FDK}}(\mathbf{x})^{\gamma}}{Z}, \qquad Z = \sum_{\mathbf{x}' \in \Omega} V_{\text{FDK}}(\mathbf{x}')^{\gamma}, \quad (4)$$

that linearly interpolates between a uniform prior and a power-shaped attenuation profile. Per-Gaussian densities are then queried from V_{FDK} at the sampled locations and per-Gaussian scales are set by nearest-neighbor spacing. The mixture form is critical: density weighting alone would collapse capacity onto the brightest bony structures, while the uniform term retains global coverage so that under-attenuating but anatomically relevant regions remain populated.

3.4. Geometry-aware Pruning (GAP)

Inside the radiative backbone, projection-error gradients are systematically larger near bone–soft-tissue boundaries, so standard gradient-driven densification keeps duplicating Gaussians around the same high-contrast surfaces. Large gradients in this regime do not always imply missing coverage: they can also reflect that an already-saturated boundary is being represented redundantly by neighboring Gaussians. Rather than adding more support points in sparse regions in the spirit of FSGS [**zhu2024fsgs**], GAP acts in the opposite direction and reclaims locally crowded Gaussians during densification.

For each Gaussian G_i we define a proximity score from its K nearest neighbors $\mathcal{N}_K(i)$,

$$s_i = \frac{1}{K} \sum_{j \in \mathcal{N}_K(i)} \|\mathbf{p}_i - \mathbf{p}_j\|_2, \quad (5)$$

and combine it with $\bar{g}_i$, the accumulated positional-gradient magnitude of G_i over the current densification cycle. A Gaussian is flagged as a redundancy candidate only when it is simultaneously locally crowded and recently inactive:

$$m_i = \mathbf{1}[s_i \geq \tau \vee \bar{g}_i \geq \delta], \quad (6)$$

i.e. $m_i = 0$ marks a prune candidate. GAP then removes a bounded fraction of the flagged Gaussians per cycle and slightly contracts the covariance of their retained neighbors to avoid creating local support holes. The dual criterion is what distinguishes GAP from purely density-based pruning: gradient-activity prevents deleting Gaussians that are still learning a boundary, while proximity prevents over-densified regions from continuously monopolizing capacity.

3.5. Adaptive Density Modulation (ADM)

Even after path-anchored initialization and redundancy control, density values across anatomy remain heterogeneous: bone exhibits higher attenuation, air-filled regions should stay near zero, and soft tissue requires smoother transitions. A single per-Gaussian gradient step has difficulty expressing this spatial heterogeneity without overshooting one regime while undershooting another. ADM therefore encodes position-dependent spatial context through K-Planes [**fridovich2022kplanes**] and uses it to predict a relative density correction, without touching Gaussian centers or covariances.

Given a position $\mathbf{x}$, we bilinearly sample three orthogonal feature planes $\{P_{uv}\}_{uv \in \{xy,xz,yz\}}$ at the corresponding 2D projections $\Pi_{uv}(\mathbf{x})$ and concatenate the results:

$$f_{uv}(\mathbf{x}) = \mathcal{B}\big(P_{uv}, \Pi_{uv}(\mathbf{x})\big), \quad uv \in \{xy, xz, yz\}, \quad (7)$$

$$F(\mathbf{x}) = \big[f_{xy}(\mathbf{x});\ f_{xz}(\mathbf{x});\ f_{yz}(\mathbf{x})\big], \quad (8)$$

where $\mathcal{B}(\cdot)$ denotes bilinear sampling. A lightweight dual-head MLP h decodes $F(\mathbf{x})$ into a density offset $\Delta\sigma(\mathbf{x})$ and a confidence gate $g(\mathbf{x}) \in [0,1]$. We center the offset by its batch mean $\overline{\Delta\sigma}$ and modulate the base density multiplicatively:

$$\rho_{\text{final}}(\mathbf{x}) = \rho_{\text{base}}(\mathbf{x}) \cdot \Big(1 + g(\mathbf{x}) \cdot \big(\Delta\sigma(\mathbf{x}) - \overline{\Delta\sigma}\big)\Big). \quad (9)$$

The plane features and the MLP parameters are trained end-to-end on the current case under the rendering loss alone, with no extra density supervision; the modulation amplitude is attenuated when fewer input views are available and is ramped in only after the backbone has reached a stable geometric estimate. Batch centering turns $\Delta\sigma$ into a relative correction rather than a global density boost,

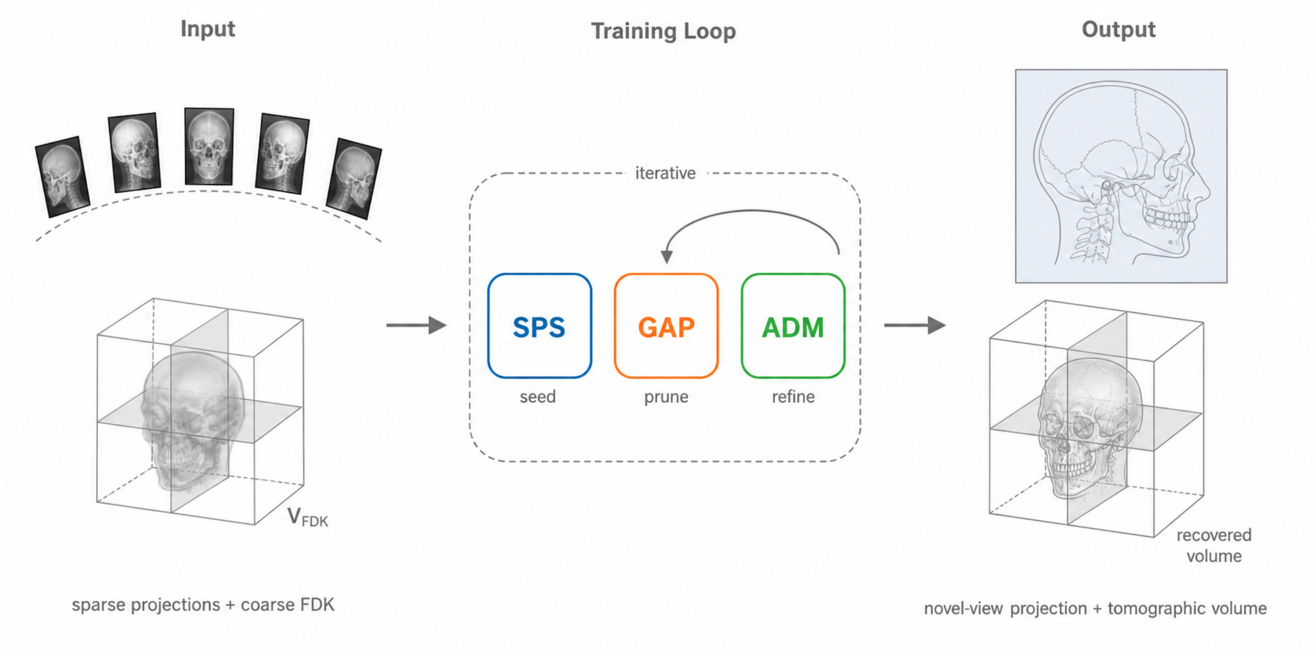

Figure 2: ***Overall framework of XRA-GS.*** *Sparse X-ray projections together with a coarse FDK reconstruction enter a radiative Gaussian backbone whose Gaussian evolution is reorganized by SPS (seed), GAP (prune), and ADM (refine). The three modules act at initialization, inside the densification loop, and during later optimization, respectively, leaving inference unchanged.*

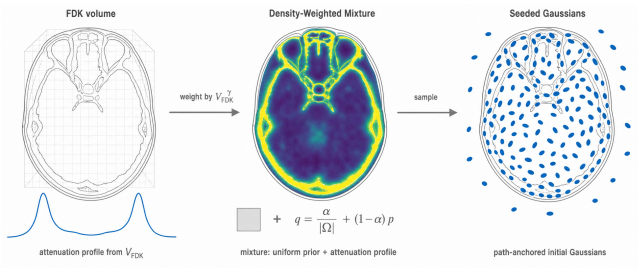

Figure 3: ***Support-Profile Seeding (SPS).*** *From the coarse FDK volume, SPS builds a density-weighted mixture distribution that interpolates between a uniform prior and the attenuation profile. Sampling from this mixture seeds Gaussians along the anatomical support before any optimization step.*

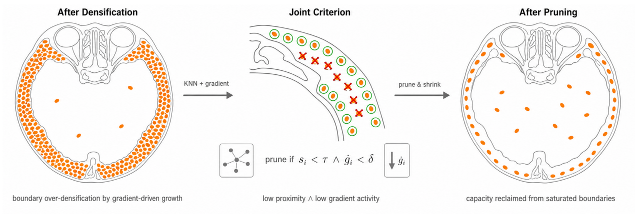

Figure 4: ***Geometry-aware Pruning (GAP).*** *After gradient-driven densification, Gaussians cluster around high-contrast boundaries. GAP flags a Gaussian as a redundancy candidate only when it is both locally crowded ($s_i < \tau$) and recently inactive ($\bar{g}_i < \delta$), so that capacity is reclaimed from saturated boundaries without removing Gaussians that are still learning structure.*

so ADM amplifies density where the attenuation support warrants finer detail and damps modulation where the evidence is weak.

3.6. Training Loss

XRA-GS is supervised end-to-end on the observed views by a single objective combining projection-domain reconstruction with regularization on the volumetric density and the K-Planes features:

$$\mathcal{L} = \mathcal{L}_1 + \lambda_d \mathcal{L}_{\text{dssim}} + \lambda_v \mathcal{L}_{\text{3DTV}} + \lambda_p \mathcal{L}_{\text{pTV}}. \tag{10}$$

The first two terms are the standard L_1 and D-SSIM losses between rendered and observed projections; $\mathcal{L}_{\text{3DTV}}$ is a total-variation prior on the volumetric density retrieved by the differentiable voxelizer of R2-Gaussian, and $\mathcal{L}_{\text{pTV}}$ is a planar total-variation prior on the K-Planes features used by ADM. The K-Planes features and the MLP parameters of ADM are jointly updated with the backbone Gaussian parameters under $\mathcal{L}$, so density modulation is learned online rather than fixed in advance.

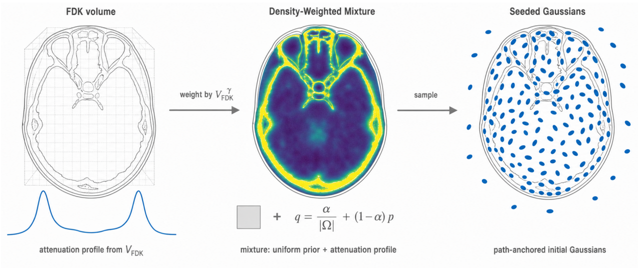

Figure 5: ***Adaptive Density Modulation (ADM).*** *For each Gaussian center, three orthogonal K-Planes feature planes are bilinearly sampled and concatenated; a dual-head MLP outputs a density offset and a confidence gate, which together apply a batch-centered relative correction* $\rho_{final} = \rho_{base}(1 + g(\Delta\sigma - \overline{\Delta\sigma}))$.

4. Experiments

4.1. Experimental Settings

Dataset. Following NAF [**guan2024naf**] and X-Gaussian [**cai2024xgaussian**], we adopt the public datasets of human organ CTs, *i.e.*, LIDC-IDRI and the open scientific visualization dataset, to evaluate our method. The test scenes include Chest, Foot, Head, Abdomen, and Pancreas. We adopt the open-source tomographic toolbox TIGRE to capture 100 projections with 3% noise in the range of $0 \sim 180^\circ$. In the sparse-view NVS task, we retain only 2, 3, or 4 projections for training, and the remaining projections are used for testing. The CT volumes are used for testing in the sparse-view CT reconstruction task.

Baselines. We compare **XRA-GS** with six state-of-the-art methods from two categories. *X-ray/CT reconstruction methods:* X-Gaussian [**cai2024xgaussian**] [ECCV'24] introduces radiative Gaussian splatting for X-ray NVS; R2-Gaussian [**hu2024r2**] [NeurIPS'24] rectifies the integration bias in 3DGS and develops a differentiable voxelizer for tomographic reconstruction; and X-Field [**xfield2025**] [NeurIPS'25] employs physically informed ellipsoids with material-aware optimization. *Natural-scene sparse-view 3DGS methods:* CoR-GS [**zhang2024cor**] [ECCV'24] introduces co-regularization to stabilize sparse-view training; DNGaussian [**li2024dn**] [CVPR'24] injects depth normalization for geometric guidance; and FSGS [**zhu2024fsgs**] [ECCV'24] augments densification with proximity-guided unpooling. All methods are evaluated under the same sparse-view protocol on the same dataset.

Metrics. We adopt peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) to evaluate the performance. Both metrics are computed on $[0,1]$-normalized projections, averaged over all held-out views of each scene and then over the five organs.

Implementation Details. Our **XRA-GS** is implemented in PyTorch and CUDA. The model is trained with the Adam optimizer for 3×10^4 iterations. SPS uses mixture coefficient $\alpha = 0.2$, power exponent $\gamma = 1.0$, and 50K initial Gaussians. GAP uses $K = 5$ nearest neighbors, proximity threshold $\tau = 0.015$, maximum pruning ratio $\beta_{\mathrm{prune}} = 2\%$, and gradient threshold $\delta = 0.0002$. ADM uses K-Planes at resolution 64×32 and activates after 15K iterations. All experiments are conducted on a single NVIDIA RTX A6000 GPU.

4.2. Comparison with State-of-the-Art Methods

Discussion on NVS Quantitative Results. Table **??** reports the quantitative comparisons between our **XRA-GS** and six state-of-the-art methods on the NVS task. We report PSNR (upper entry in each cell) and SSIM (lower entry) of all methods on all scenes under the 2/3/4-view sparse settings. The natural-scene sparse-view 3DGS methods (CoR-GS, DNGaussian, FSGS) struggle under the X-ray imaging regime, since their densification strategies and geometric priors are designed for visible-light surface rendering rather than line-integral attenuation. Among the X-ray-specific methods, X-Gaussian achieves reasonable performance but remains limited by the lack of integration bias correction. R2-Gaussian provides strong baseline results through its rectified rasterization and differentiable voxelizer. Our **XRA-GS** achieves the best average performance across all sparsity levels. The most pronounced margin occurs at the 3-view setting, consistent with our central argument: capacity misallocation is most directly addressable when projection evidence is sufficient to guide redistribution yet too limited for standard densification to converge to a spatially consistent solution. X-Field shows competitive results in the human organ reconstruction setting, but its material-based optimization faces challenges under extremely sparse views where material boundaries are ambiguous.

Discussion on NVS Qualitative Results. Figure **??** depicts the qualitative comparisons of NVS on four representative organs (Chest, Head, Foot, and Pancreas) under the 2-view, 3-view, and 4-view settings. Under highly sparse view settings, all methods face challenges from limited input information. DNGaussian produces noticeable blurry artifacts due to its depth-normalization scheme being misaligned with the X-ray transmission model. R2-Gaussian reconstructs the overall anatomical structure better than the natural-scene baselines, but still exhibits boundary over-accumulation artifacts, particularly visible as streak patterns near high-contrast bone–soft-tissue interfaces. X-Field produces competitive results with reduced artifacts in organ interiors, though under the 2-view setting it introduces wave-pattern artifacts in the head scene. Our **XRA-GS** effectively mitigates these artifacts by reallocating Gaussian capacity from over-densified boundaries to interior attenuation-relevant structures. In particular, **XRA-GS** recovers smoother soft-tissue regions in the Chest, sharper bone details in the Foot, and cleaner background in the Head, while maintaining structural consistency across all three view budgets.

4.3. Ablation Study

Component Analysis. We conduct a break-down ablation experiment to study the effect of each proposed module. We adopt the R2-Gaussian backbone as the baseline model. The results are listed in Table **??**. We can observe from the table: **(i)** When adding SPS for initialization, the performance yields improvement especially under the most under-determined 2-view regime, establishing that path-anchored initialization from the coarse FDK attenuation support provides a meaningful head start for optimization. **(ii)** Adding GAP delivers the largest PSNR increment by

Table 1: *Quantitative results on the novel view synthesis task. We colorize the best, second-best, and third-best numbers. PSNR (upper) and SSIM (lower) are reported in each cell. All methods are evaluated under the unified five-organ protocol.*

Method	Chest			Head			Abdomen			Foot			Pancreas			Average		
	2v	3v	4v	2v	3v	4v	2v	3v	4v	2v	3v	4v	2v	3v	4v	2v	3v	4v
Natural-scene sparse-view 3DGS methods																		
CoR-GS [**zhang2024cor**] [ECCV'24]	14.96	18.55	20.29	16.24	20.36	23.40	18.50	22.45	23.03	20.17	25.53	26.83	16.15	23.25	23.96	17.20	22.03	23.50
	0.5754	0.7077	0.7636	0.7993	0.8644	0.9088	0.8645	0.9089	0.9269	0.6913	0.8511	0.8767	0.8088	0.8974	0.9084	0.7479	0.8459	0.8769
DNGaussian [**li2024dn**] [CVPR'24]	18.84	20.55	20.99	15.72	18.04	16.24	14.30	16.02	15.76	19.33	23.51	24.72	19.06	22.43	25.47	17.45	20.11	20.64
	0.6675	0.7500	0.7791	0.7296	0.8097	0.7924	0.7663	0.8258	0.8486	0.7801	0.8569	0.8666	0.8082	0.8848	0.8982	0.7503	0.8254	0.8370
FSGS [**zhu2024fsgs**] [ECCV'24]	17.86	19.99	21.03	19.43	21.53	24.91	18.75	23.34	23.63	22.17	25.37	27.08	23.03	25.27	26.22	20.25	23.10	24.57
	0.6451	0.7429	0.7729	0.8334	0.8732	0.9105	0.8580	0.9068	0.9276	0.6843	0.8500	0.8760	0.8564	0.9031	0.9148	0.7754	0.8552	0.8804
X-ray/CT reconstruction methods																		
X-Gaussian [**cai2024xgaussian**] [ECCV'24]	17.79	20.29	20.90	19.62	21.52	24.46	18.93	23.54	23.85	22.34	25.48	26.92	23.21	25.12	25.55	20.38	23.19	24.34
	0.6488	0.7427	0.7689	0.8347	0.8697	0.9079	0.8595	0.9077	0.9257	0.6882	0.8481	0.8740	0.8502	0.9006	0.9124	0.7763	0.8538	0.8778
R2-Gaussian [**hu2024r2**] [NeurIPS'24]	21.05	26.22	25.47	23.54	26.60	28.66	24.85	29.26	30.56	19.39	28.48	30.04	17.83	28.61	30.90	21.33	27.83	29.13
	0.7110	0.8367	0.8546	0.8718	0.9223	0.9550	0.9028	0.9363	0.9617	0.6722	0.8977	0.9130	0.8122	0.9199	0.9360	0.7940	0.9026	0.9241
X-Field [**xfield2025**] [NeurIPS'25]	19.57	23.94	24.45	22.49	25.06	26.78	22.20	24.87	27.19	20.45	26.03	27.25	18.12	24.94	26.36	20.57	24.97	26.41
	0.6843	0.8090	0.8255	0.8413	0.8878	0.9089	0.7190	0.8632	0.9103	0.6522	0.8405	0.8556	0.6871	0.8474	0.9004	0.7168	0.8496	0.8801
XRA-GS (Ours)	20.97	26.63	25.62	23.97	26.55	28.76	25.26	29.55	30.90	19.44	28.34	29.84	17.97	29.37	30.84	21.52	28.09	29.19
	0.7111	0.8390	0.8523	0.8700	0.9179	0.9536	0.9068	0.9374	0.9627	0.6785	0.8993	0.9141	0.8147	0.9247	0.9361	0.7962	0.9037	0.9238

reclaiming capacity from over-densified boundary regions, confirming that geometry-aware pruning is the primary mechanism for correcting the capacity misallocation inherited from visible-light densification. **(iii)** Adding ADM provides further refinement through position-dependent density correction, particularly benefiting scenes with heterogeneous attenuation profiles. The full **XRA-GS** pipeline, combining all three modules, achieves the best performance across all view counts.

Parameter Analysis. We further investigate the sensitivity of one representative hyperparameter from each module in Table **??**. **(i)** The SPS mixture coefficient α balances the FDK-derived attenuation support against random initialization; $\alpha = 0.2$ offers sufficient path-anchored guidance without over-constraining the initial distribution. **(ii)** The GAP proximity threshold τ controls how aggressively crowded boundary Gaussians are pruned; a larger τ retains more crowded Gaussians and reduces the pruning effect, while a smaller τ prunes more aggressively and risks removing useful Gaussians near legitimate boundary structures. **(iii)** The ADM activation iteration determines when density modulation is introduced; activating too early interferes with the backbone's geometric convergence, while activating too late limits the modulation's contribution. The default configuration ($\alpha = 0.2$, $\tau = 0.015$, ADM activation at 15K) achieves robust performance across the swept range, demonstrating that **XRA-GS** is not sensitive to individual hyperparameter choices.

Table 2: *Ablation results with our choices in **bold**. We colorize the best, second-best, and third-best numbers.*

Configuration	PSNR↑	SSIM↑	Time↓	#Gauss.
Component analysis (5-organ avg, 3-view)				
Baseline (R2-Gaussian)	–	–	–	–
B + SPS	–	–	–	–
B + GAP	–	–	–	–
B + ADM	–	–	–	–
Full XRA-GS	–	–	–	–
SPS mixture coefficient α				
$\alpha = 0.0$	–	–	–	–
$\alpha = 0.1$	–	–	–	–
$\boldsymbol{\alpha = 0.2}$	–	–	–	–
$\alpha = 0.5$	–	–	–	–
$\alpha = 1.0$	–	–	–	–
GAP proximity threshold τ				
$\tau = 0.005$	–	–	–	–
$\tau = 0.010$	–	–	–	–
$\boldsymbol{\tau = 0.015}$	–	–	–	–
$\tau = 0.020$	–	–	–	–
$\tau = 0.030$	–	–	–	–
ADM activation iteration				
Iter = 5K	–	–	–	–
Iter = 10K	–	–	–	–
Iter = 15K	–	–	–	–
Iter = 20K	–	–	–	–
Iter = 25K	–	–	–	–

Placeholder: `assets/fig/experiment_fig_qual_nvs.png`

Layout: 3 blocks (2-view / 3-view / 4-view), each block contains:
4 rows (Chest / Head / Foot / Pancreas) $\times$ 5 columns (GT / DNGaussian / R2-Gaussian/ X-Field / **XRA-GS**)

Each image with zoomed-in patch; PSNR annotated below each image.
Total: $3 \times 4 \times 5 = 60$ image cells.

Figure 6: ***Qualitative comparison of NVS.*** *We present visual examples of rendered X-ray projections across four organs (Chest, Head, Foot, Pancreas) under three sparse-view settings (2, 3, and 4 input views). Columns: GT, DNGaussian, R2-Gaussian, X-Field, and* ***XRA-GS*** *(Ours). Our method produces clearer structural details, smoother textures in homogeneous regions, and fewer artifacts near anatomical boundaries. Red boxes highlight regions of interest for detailed comparison.*

5. Conclusion

XRA-GS demonstrates that visible-light densification rules impose a systematic capacity-allocation failure in transmissive X-ray imaging that persists even after the rendering equation is physically corrected. By intervening at three stages of Gaussian evolution—path-anchored initialization (SPS), boundary redundancy recovery (GAP), and position-dependent local refinement (ADM)—the framework redirects representation capacity from over-densified anatomical boundaries toward the interior attenuation-relevant structures that conventional gradient-driven growth neglects. Across a five-organ benchmark with 2/3/4 input views against six representative baselines, **XRA-GS** achieves the best average PSNR2D at every sparsity level, with SSIM2D reported for methods whose same-protocol logs are available. The strongest PSNR2D improvement appears in the moderately sparse 3-view regime, where sufficient projection evidence allows capacity reallocation to take effect.